



Faculty of Informatics and Computer Science

Software Engineering

Project Title

ComicMind: AI Comic Book Creator Webservice

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December 2025

Abstract

Comics creators who use AI image-generation technologies now demand platforms which enable them to create coherent visual narratives instead of generating single images. The current platforms fail to maintain consistent character appearances and continuous storytelling and editable dialogue, which restricts their value for creators who need to develop complete comics. The current tools for comic creation face two major problems because they either embed text directly into artwork, which hinders translation and revision, or they need users to purchase multiple paid services and perform complicated manual editing. The current platform lacks essential features which make it difficult for users to create comics because it does not support continuous storytelling, and users need to perform complex editing tasks.

his project “ComicMind: AI Comic Book Creator Webservice” develops an end-to-end AI-powered webservice which helps the generation of six-panel comics through automated processes that maintain both narrative structure and visual consistency. The ComicMind webservice starts by developing user-premises into organised story beats, which produce scenes and dialogue that advance the story in a logical way. ComicMind generates visual prompts through deterministic character control to achieve uniform character appearances throughout all panels. ComicMind also generates text-free images through a backup system of AI providers, which maintains reliable output when any service experiences failure. Users can edit dialogue and narration through Scalable Vector Graphics (SVG)-based speech bubbles, which provide complete control for translation, accessibility, and stylistic adjustments.

ComicMind serves users who lack artistic skills and includes educators, content producers, and rapid prototypers who need a versatile tool to create comics and storyboards. It provides users with editable text layers, organised panel arrangements, and uniform character designs, which enhance their ability to understand stories. ComicMind enables users to store dialogue independently from images, which enables multilingual output and supports accessibility features and localisation. The user dashboard enables creators to handle their comics through viewing, deleting, and exporting options, while content safety filters maintain proper usage.

ComicMind uses a contemporary web development framework which combines React for frontend development with Node.js and Express for backend development and MongoDB for storing comics and user data and metadata. The system includes multiple security measures and

AI provider tracking and a defined workflow for generating stories and creating prompts and placing speech bubbles ComicMind will deliver a complete comic generation service which solves current platform weaknesses through its ability to create coherent content with user-friendly customization options. It establishes comic creation accessibility through its combination of AI storytelling assistance with dependable generation systems and complete output editing capabilities.

Attestation & Turnitin Report

I understand the nature of plagiarism, and I am aware of the University's policy on this.

I certify that this report reports original work by me during my University project except for the following (*adjust according to the circumstances*):

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Acknowledgements

Acknowledge anyone who has helped you in your work such as your supervisor, technical support staff, fellow students or external organisations. Acknowledge the source of any work that is not your own.

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1 Introduction

1.1 Overview

This project “Comic Mind: AI Comic Book Creator Webservice” aims to support the development of AI-generated comics. The target is to construct six-panel stories that maintain the characters consistency and flow of a story. Existing tools only make pictures without character consistency and dialogue. So far, existing tools have been used to put text on top of pictures. These tools either do not offer character consistency and text-editing options or provide these features as paid services. This added process is what makes making comics creation difficult and slow. With these constraints in place, the proposed service provides an easy approach for visual storytelling. ComicMind is a solution for those users who would like an interface to take an idea and turn it into lined panels. It helps them easily practice visual storytelling. ComicMind is generic as it supports all types of users: non-artists, teachers, content creators and those who need storyboards or visual material. It provides dialogue layers, character consistency. Users of ComicMind can construct comics that are easily changeable or reusable.

1.2 Problem Statement

AI image-generation tools have surged in popularity, but most fail to produce complete, engaging comics. Current platforms typically generate isolated images without narrative cohesion or consistent characters. Common limitations include:

- One-off images: No support for multi-panel storytelling or maintaining continuity between scenes.
- Weak narratives: Generic, repetitive dialogue and formulaic plots typical of “AI slop” writing.
- Inaccessible text: Speech and dialogue are often embedded directly into the image, preventing editing, translation, or localization.
- Character inconsistency: Visual discrepancies across panels make the story feel fragmented and unprofessional.
- High cost and complexity: Many workflows rely on multiple paid APIs or manual post-processing in tools like Photoshop.

This project aims to address these issues by building an end-to-end AI-powered comic generation pipeline that prioritizes story-first design and character consistency. The service supports:

- Automated generation of six-panel story structures.
- Deterministic character control for consistent appearances across scenes.
- Clean, text-free images with SVG-based speech bubbles intelligently placed during post-processing.
- Error handling to ensure reliable output even when image generation fails.

ComicMind aspire to support underserved Target Audience such as: Non-artists who want to tell personal or educational stories (students, educators, content creators), users in low-resource languages/locales needing editable text overlays and accessibility, and rapid prototypers (indie devs/writers) who need coherent storyboards with consistent characters.

ComicMind should fulfill this audience Needs and Preferences which basically include: visual narrative comprehension improves with consistent character depiction and clear panel structure, editable text layers (not baked into images) improve accessibility, localization, and iteration speed, and some design implication such as: Prioritizing story quality, character consistency, and rapid iteration with safe fallbacks.

1.3 Scope and Objectives

The objective of ComicMind this project is to create an end-to-end AI-powered platform for generating six-panel comics that maintain narrative coherence, character consistency. The system aims to solve the common problems found in existing AI-based comic tools, such as inconsistent character appearances, weak and repetitive storytelling, text embedded directly inside images, and the need for multiple external tools to assemble a full comic.

The platform will allow users to enter a simple idea and transform it into a structured story with clear scenes, dialogue, and narration. It will then generate clean images for each panel, place speech bubbles, and allow users to preview, save, and export their comics in multiple formats. In addition, the platform system will include user accounts, a gallery for managing previous comics, and content-safety mechanisms to ensure appropriate usage. The platform will be

fully responsive to provide a smooth experience on desktop. Both non-artists and creative users such as students, educators, and content creators will benefit from the system, as it simplifies comic creation, improves accessibility, and removes the need for manual graphic design skills.

Scope:

- Design a simple and user-friendly interface where users can enter a title, idea, and visual style to generate a complete six-panel comics.
- Implement AI-based story generation that produces structured story, scene descriptions, dialogue, and narration while maintaining narrative clarity and consistency.
- Integrate character-consistent image generation using seeded prompts.
- Create speech-bubble overlays using SVG so that dialogue is not embedded inside images.
- Construct a database of user accounts and comic projects, allowing users to save, view, delete, and re-open previously generated comics.
- Develop a comic preview and export module that displays all six panels clearly and supports exporting comics as PNG or PDF.
- Implement content-safety features to detect inappropriate prompts and provide user warnings, ensuring responsible use of the platform.
- Build a responsive website that works smoothly across desktops.

1.4 Report Organization (Structure)

This report is divided into several chapters. Chapter 1 explains what the project is about, the problem it tries to solve, the goals of the system, and how the rest of the report is organized. It also gives a brief idea of the methodology and the work plan that were followed while developing ComicMind.

Chapter 2 talks about the related work and background. It goes through the tools and research that are similar to this project and explains their limitations, which helped show why ComicMind was needed.

Chapter 3 describes the solution that was built. It explains how the system works, how the story and image generation pipeline functions, and the main features that ComicMind provides. The chapter also includes the functional and non-functional requirements.

Chapter 4 gives a summary of what has been done and mentions the changes that happened during the project.

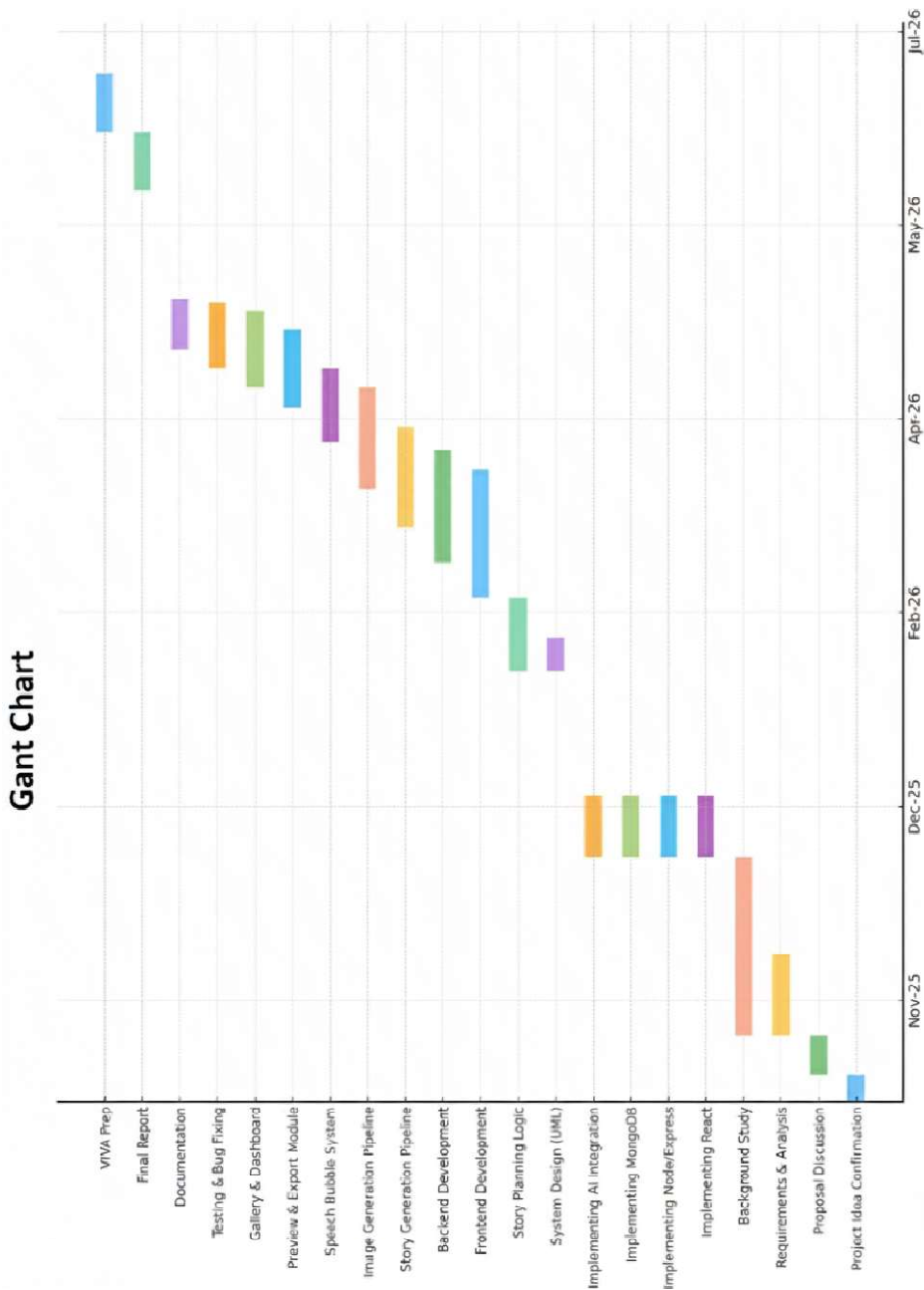
The report ends with the references.

1.5 Work Methodology

ComicMind started with the research and planning phase, where the problem of AI comic generation was explored and the limitations of existing tools were reviewed. Similar services were surveyed and analyzed to understand what they lack and how ComicMind can fill these gaps. After confirming the idea and completing the requirements and background study, the project pipeline was planned and divided into its main components: frontend, backend, database, and the AI pipelines for story and image generation. The technologies to be used were also decided during this stage, such as React for the frontend, Node/Express for the backend, MongoDB for storing user and comic data, and the OpenAI API for generating stories and images.

Once the plan was clear, development started by building the core system features step by step. This included implementing the user interface, backend routes, database models, story generation pipeline, image generation pipeline, the speech bubble system, as well as the preview, export, and gallery pages. Testing and bug fixing took place continuously from the first day of development to ensure that each part of ComicMind worked as expected. The final stage focused on documenting the system, writing the final report, and preparing for the VIVA.

1.6 Work Plan (Gantt Chart)



2 Related Work (State-of-The-Art)

2.1 Background

Recent research on human–AI co-creation highlights how digital comic and storytelling tools have evolved from manual editors into structured, AI-assisted creative environments. Early systems required users to create every scene themselves, but modern platforms now support idea expansion, story phase structuring, and visual generation with users guiding the process. Studies on author–AI collaboration show that creators appreciate step-by-step workflows where they provide a short idea, and the AI expands it into scenes that they can refine, which significantly reduces brainstorming time and helps overcome writer’s block [1]. Similar findings [2] demonstrates how preview panels, variations, and lightweight AI suggestions help authors maintain control while benefiting from efficient ideation and visual creativity. These systems mark a shift from static template-based comic tools (e.g., Pixton, Canva) toward dynamic, adaptive, AI-supported interfaces that offer transparency and user-driven refinement throughout the process. Research on narrative world-building further confirms that authors prefer structured guidance—such as generating story phases, conflicts, and resolutions—rather than full automation, because structured suggestions preserve coherence and help users understand and influence the evolving narrative [3].

Across all three studies [1, 2 and 3], AI systems are most successful when they support creativity rather than replace it. Users consistently value workflows that break the task into clear steps, maintain narrative structure, and provide multiple variations without overwhelming the interface. These approaches reduce cognitive load, speed up creation, and keep users engaged.

studies highlighted that AI systems that support creativity without removing user control receive the most positive feedback. Users appreciate step-by-step workflows, structured story generation, quick idea expansion, and interfaces that present options clearly [2]. These approaches reduce cognitive load and speed up the creation process, making AI tools more engaging. However, studies also reported many tools shortcomings such as loss of narrative or visual consistency, excessive automation that replaces rather than supports the author, confusing or cluttered interfaces, and unexpected AI changes with no explanation. These shortcomings have lead to the reduce of trust, increase of user frustration, and cause users to abandon the tool [2].

Research recommended some best practices for Human–AI co-creation systems that they should maintain a predictable structure for storytelling, give users the ability to correct or retry specific outputs, keep interfaces simple and focused, make AI behavior transparent, and support, rather than override, human creativity.

2.2 Literature Survey

1) Pixton¹

Pixton is a comic generation platform that provide many features to the users such as template-based comic maker, avatar creation, classroom management (Edu), speech bubbles, back-grounds, asset library, sharing. Pixton is designed to serve teacher and student roles in Edu version. It offers very fast procedure to assemble a comic; robust classroom features; and clear bubble tools. However it has limited AI story generation capability. The character consistency relies on templates, not generative pipeline; so less suited for custom multi-panel AI narratives.

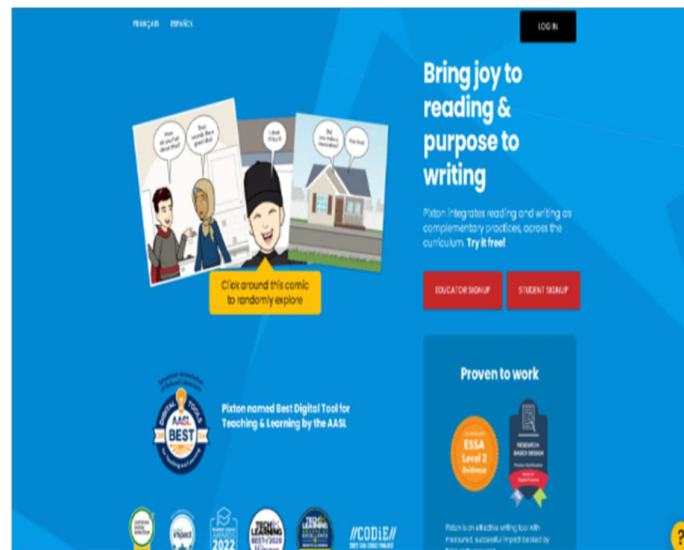


Figure 1: Home page of Pixton comic creation website

¹ <https://www.pixton.com/>

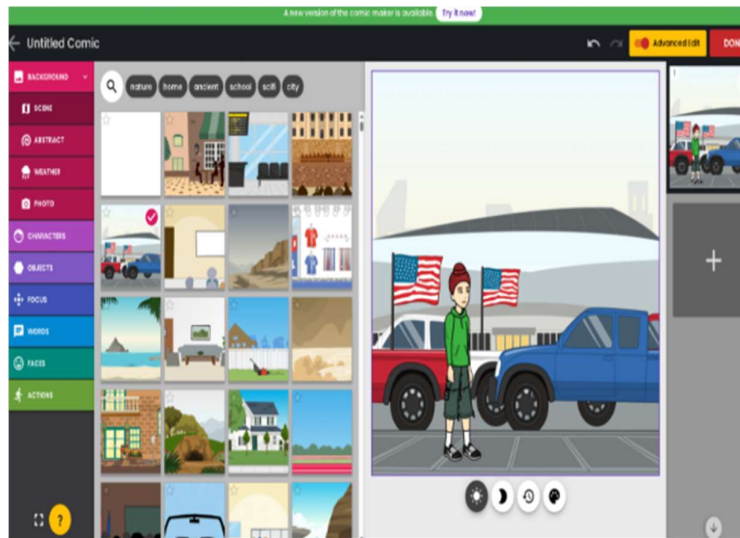


Figure 2: Creating new comic using Pixton website templates

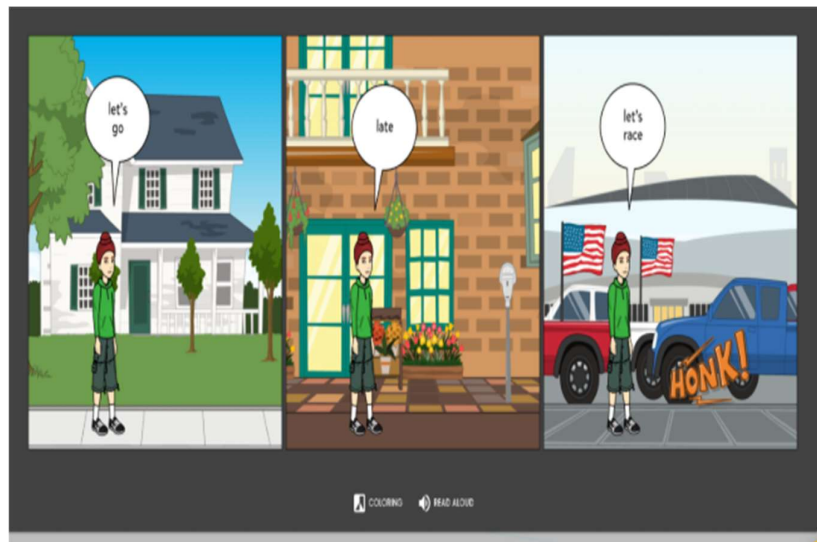


Figure 3: Example of 3-panel comic created using Pixton comic creation website

2) Canva Comic Strip Maker²

Canva Comic Strip Maker is a template-based design platform that provides drag-and-drop editing, speech bubbles, font controls, layouts, and export options, supported by light AI features such as Magic Write and Magic Design. It is designed for individual creators and collaborative

² <https://www.canva.com/create/comic-strips/>

teams who need fast visual assembly and polished layouts. The editor offers refined typography, flexible positioning, and easy export workflows. However, Canva does not function as an end-to-end AI comic generator; users must manually maintain multi-panel consistency, and AI-generated images often embed text directly into artwork unless external overlays are added, making the tool less suitable for structured AI-driven storytelling or character-consistent narratives.

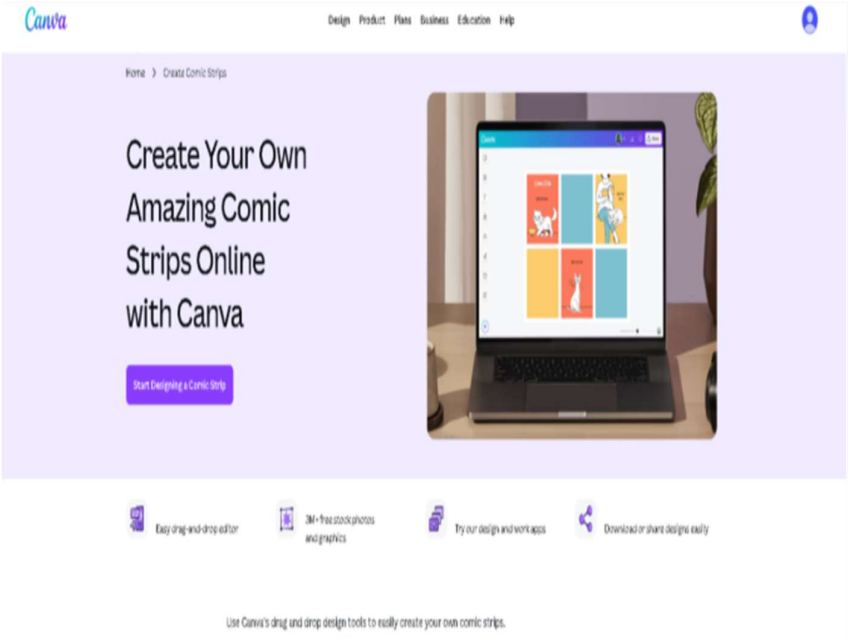


Figure 4: Home page of Canva comic creation website

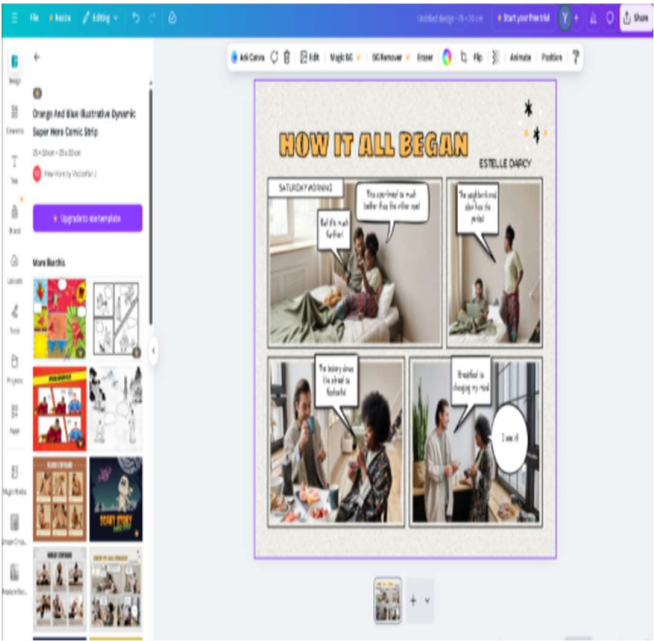


Figure 5: Canva comic editor showing a multi-panel comic layout with template options and edit-ing features

3) Midjourney³

Midjourney is a high-quality AI image-generation service offering stylistic variation, reference-based prompting, and detailed visual control, used by creators ranging from hobbyists to professional designers. It excels at producing visually striking images with strong artistic direction. However, it lacks a native comic-creation pipeline, provides no guarantees for character consistency, and offers no dialogue overlays or panel-based structure; creators must manually compile panels using external editors. Costs increase with usage, and the absence of built-in narrative tools or multi-panel orchestration makes it unsuitable for generating coherent AI comics without substantial manual work.

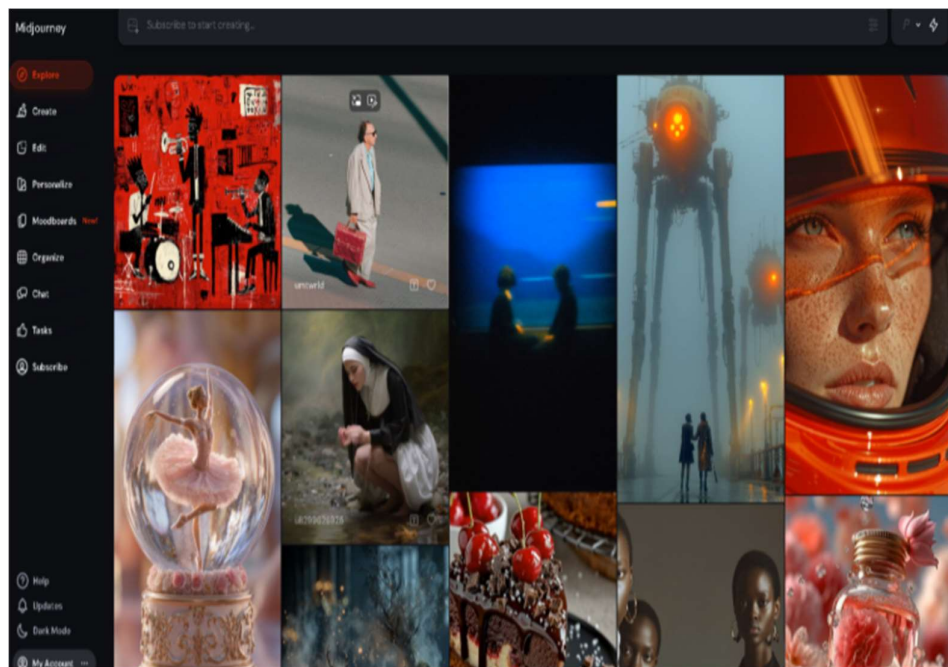


Figure 6: Home page of Midjourney showing the image gallery and available navigation options for exploring and creating AI-generated artwork

4) NovelAI Diffusion⁴

NovelAI Diffusion is a text-to-image platform known for its strong anime-style output, tag-based control system, and local image editing tools, available through subscription tiers. Users benefit from flexible control over style, lighting, and character appearance. Despite its strengths in generating attractive illustrations, the platform lacks any structured comic workflow—there

³ <https://www.midjourney.com>

⁴ <https://novelai.net>

are no story beats, no built-in panel assembly, and no speech-bubble overlays. As a result, users must manually construct narratives and manage consistency across panels, limiting its effectiveness as a tool for multi-panel AI comics.

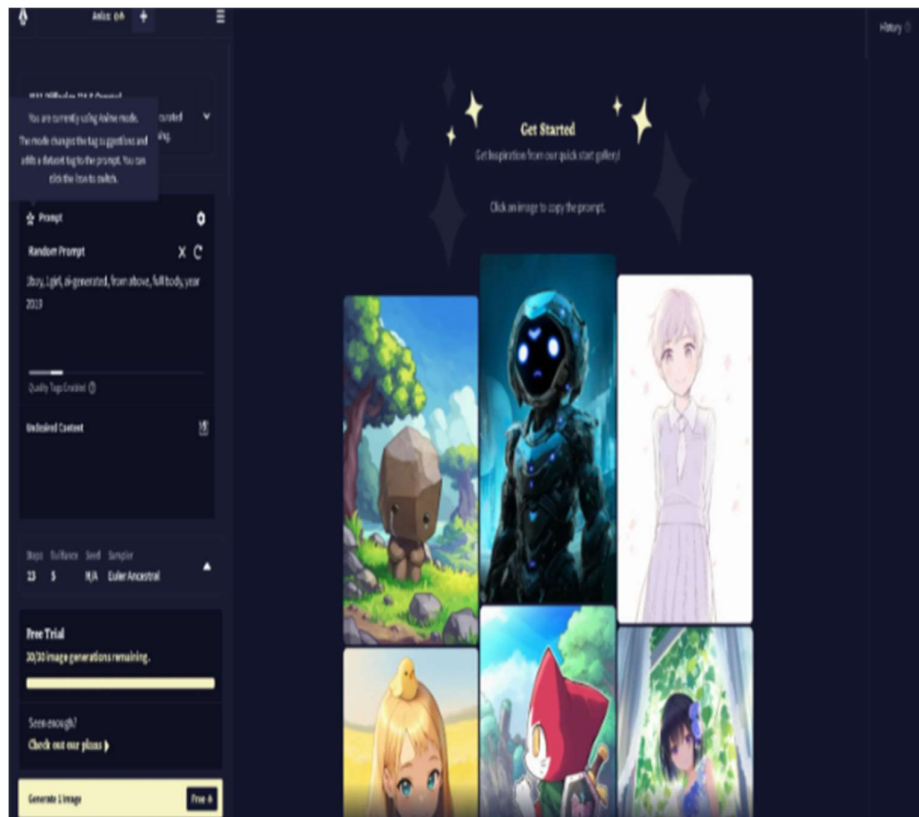


Figure 7: NovelAI Diffusion interface showing the prompt input panel and a gallery of AI-generated sample images

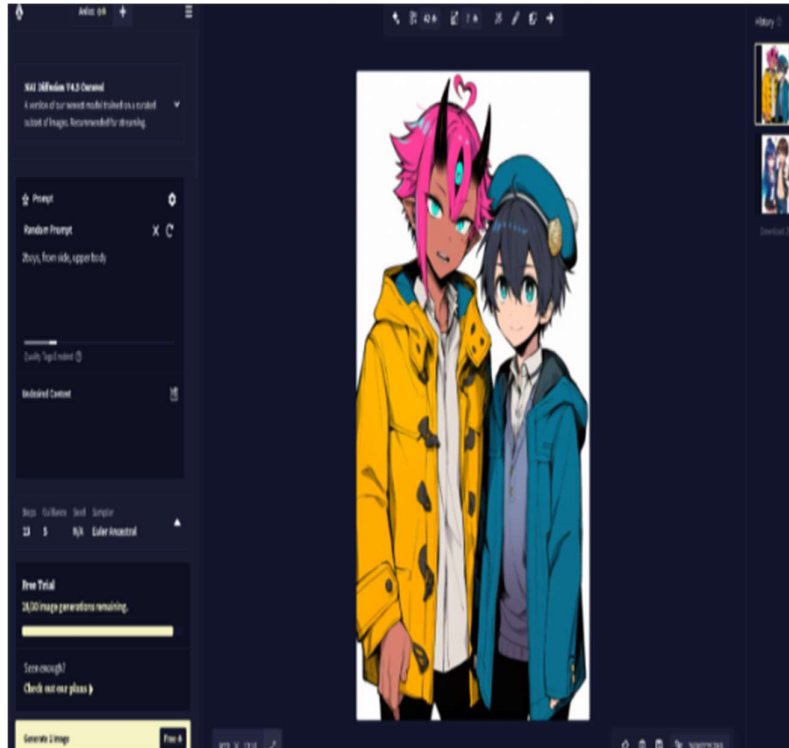


Figure 8: NovelAI Diffusion editor displaying a generated character illustration along with the prompt settings and image history panel

5) Stable Diffusion Web UI⁵

Stable Diffusion WebUI is a locally hosted image-generation interface that provides extensive control over models, prompts, seeds, ControlNet, and inpainting, enabling advanced creators to craft consistent images and scripted workflows. It supports reproducibility and can maintain visual continuity through seed management. While powerful for technical users, it is complex to operate and lacks beginner-friendly workflows, comic-specific features, panel layouts, or automatic dialogue overlays. Users must assemble multi-panel stories manually, making it unsuitable for streamlined comic generation despite its flexibility and control.

⁵ [Stable Diffusion WebUI Online](#) | [Stable Diffusion Online](#)

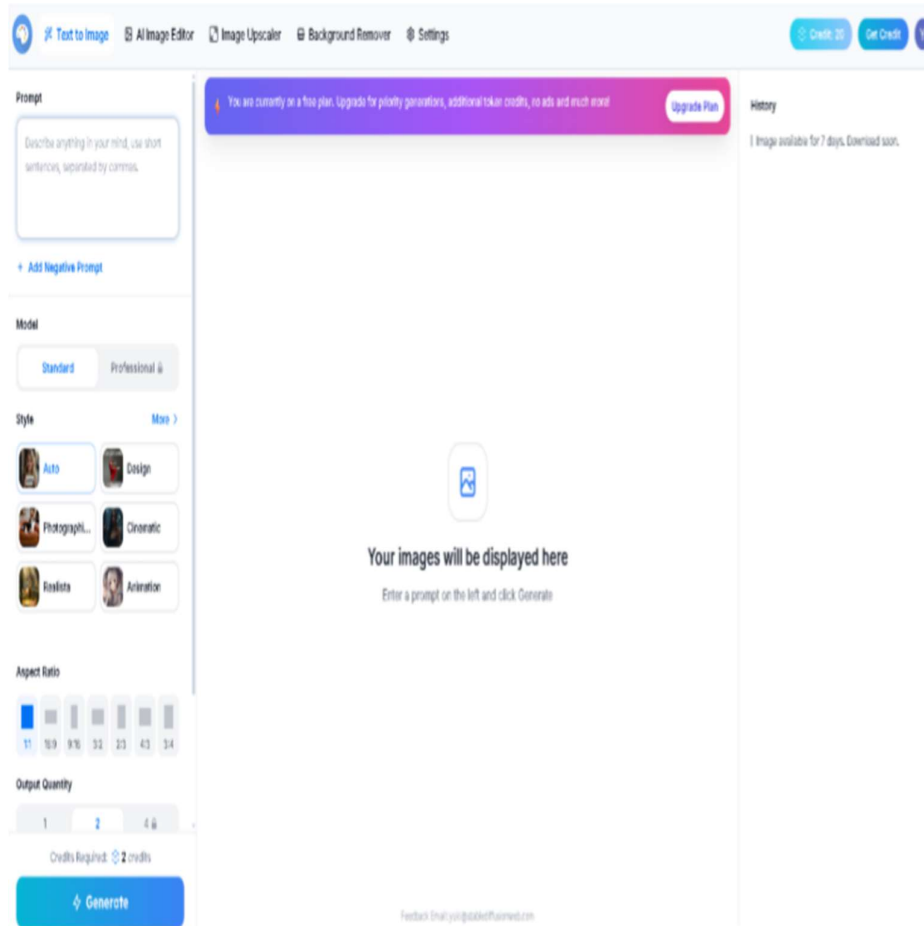


Figure 9: Stable Diffusion web interface showing the text-to-image prompt panel, style options, and generation settings

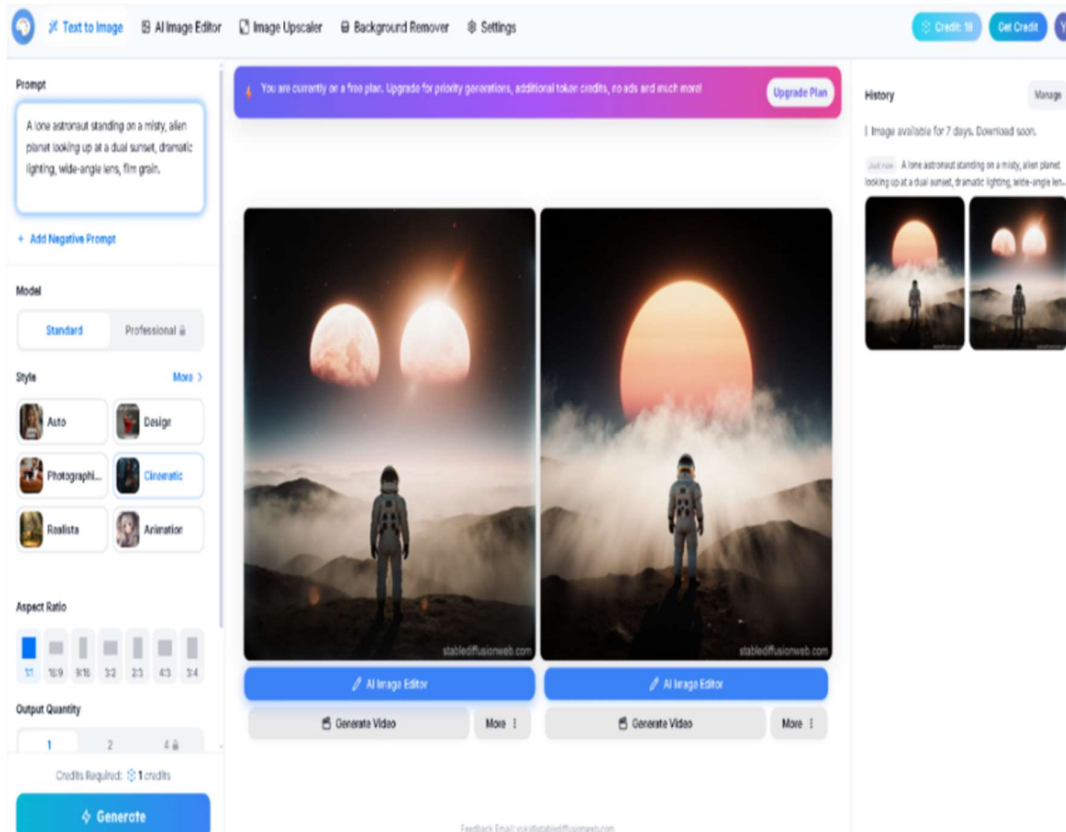


Figure 10: Stable Diffusion output page displaying generated images alongside the prompt settings and history panel

6) Storyboarder⁶ (Wonder Unit)

Storyboarder is a visual planning tool designed for rapid scene sketching, shot composition, and timing control, with the ability to export projects to editing suites such as Adobe Premiere or Final Cut Pro. It supports creators who need to conceptualize scenes quickly and iterate through camera positions and narrative pacing. It's simple sketch-based interface encourages experimentation and fast ideation. However, it includes no AI image-generation, no dialogue tools, and no comic-focused functionality such as panel rendering or speech-bubble placement, making it useful for pre-production but not for generating completed AI-driven comics.

⁶ <https://wonderunit.com/storyboarder/>

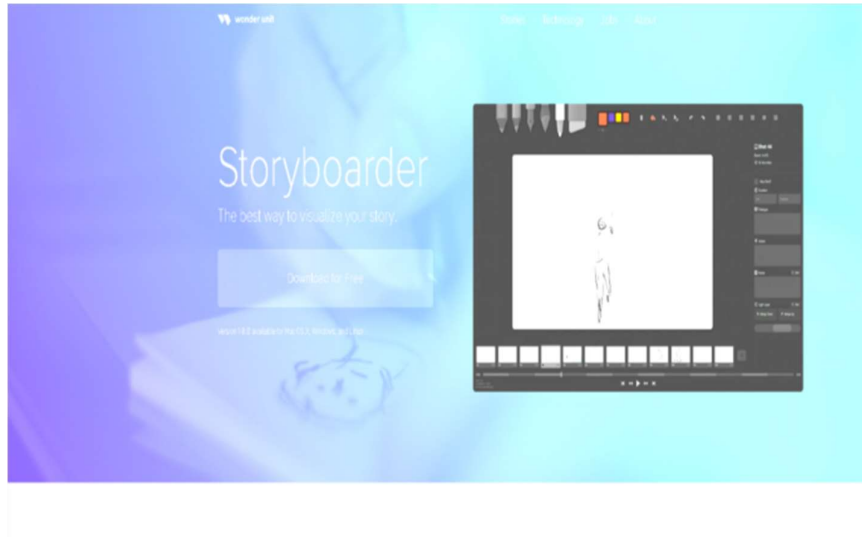


Figure 11: Storyboarder home page displaying the application's interface overview and down-load option

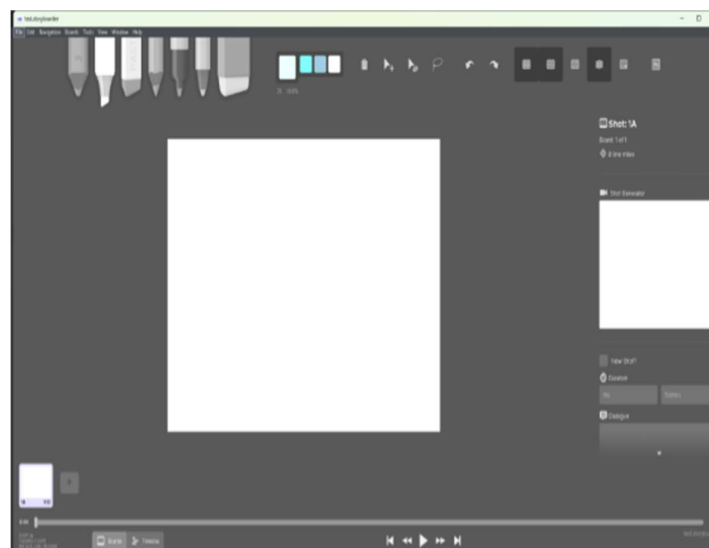


Figure 12: Storyboarder drawing interface showing the sketching tools, canvas area, and storyboard timeline

7) GoEnhance AI Comic Generator⁷

GoEnhance AI offers a comic-style image generator that converts text prompts into stylized panels with varied artistic modes, expressive characters, and dynamic environments. It integrates into GoEnhance's larger suite of creative tools, making it convenient for users who produce other digital assets on the platform. Visual quality is strong, and the system supports a range of moods and styles. However, the tool is not optimized for full comic workflows; it lacks

⁷ <https://www.goenhance.ai/tools/ai-comic-generator>

story-phase structuring, panel-to-panel continuity, dialogue overlays, and multi-panel editing. As a result, users must assemble narratives manually, limiting its effectiveness for structured AI storytelling.

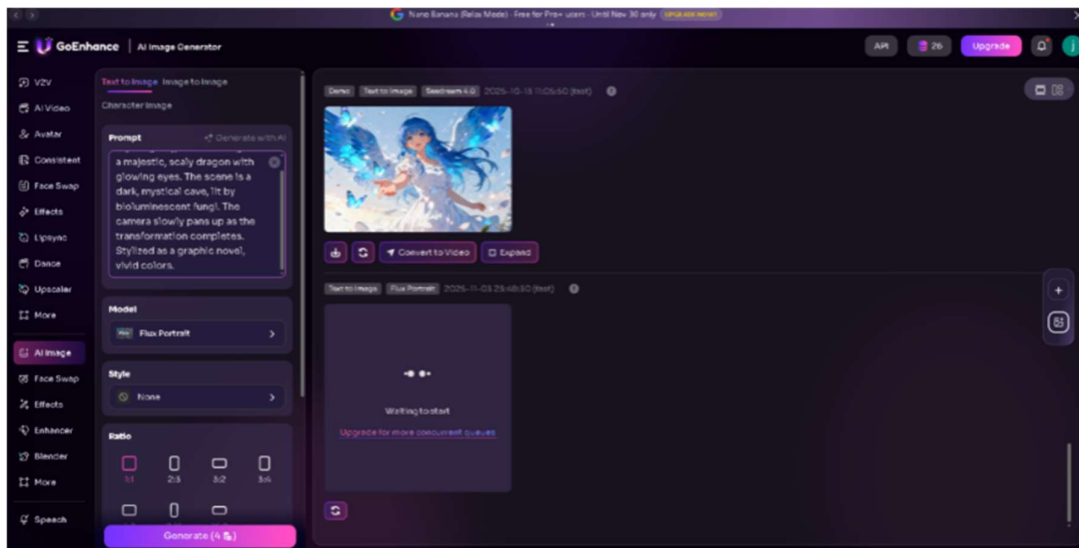


Figure 13: GoEnhance AI image generator interface showing the text-to-image prompt panel, model selection options, and generated preview

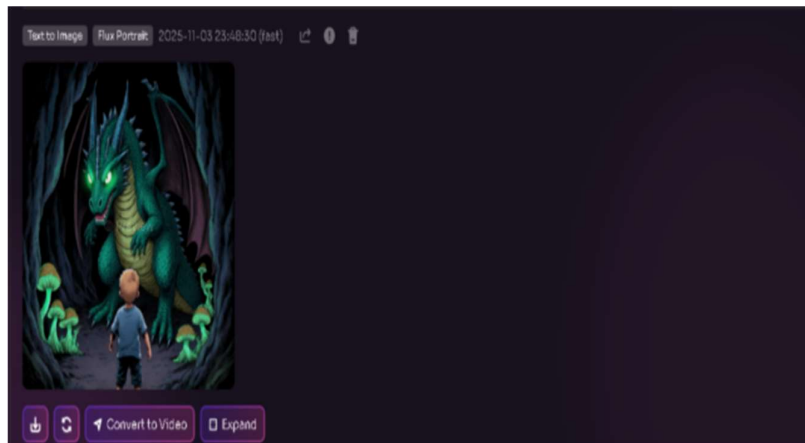


Figure 14: GoEnhance AI output display presenting the finalized generated image with available export and editing options

2.3 Analysis of the Related Work

The existing comic-creation and image-generation platforms each offer useful strengths but also share significant limitations. Pixton and Canva provide fast assembly, polished layouts, and clear speech-bubble tools, but neither offers end-to-end AI storytelling or automated multi-panel consistency. Midjourney and NovelAI deliver high-quality, stylistically rich images, yet both require manual assembly, provide no structural narrative support, and cannot maintain character consistency across panels. Stable Diffusion WebUI offers powerful control, reproducibility, and advanced editing tools, but it is complex, not beginner-friendly, and lacks any built-in comic workflow. Storyboarder excels at quick scene ideation and visual planning but includes no AI generation or dialogue features. GoEnhance produces visually appealing panels with varied styles, yet it does not support multi-panel editing, dialogue overlays, or story continuity. Overall, while each tool excels in isolated strengths—speed, layout control, image quality, or planning—none provides a complete, coherent, or automated pipeline for generating consistent multi-panel AI comics. The table below shows the feature comparison between the existing platforms and ComicMind.

Table 1: Features comparison of surveyed comic creation services and ComicMind

Function / Feature	Pixton	Canva	Midjourney	Stable Diffusion Web UI	NovelAI Diffusion	Storyboarder	GoEnhance AI	ComicMind
End-to-end 6-panel generation	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Will do
Character consistency	Partial	Doesn't exist	Doesn't exist	Partial	Partial	Doesn't exist	Partial	Will do
Speech bubbles (auto placed)	Exists	Exists	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Will do
Text-free images	Partial	Partial	Doesn't exist	Exists	Exists	Exists	Exists	Will do
Story quality control	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Will do

Provider fallback chain	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Won't do
Accessibility / localization	Exists	Exists	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Doesn't exist	Won't do
User login / sign-up	Exists	Exists	Exists	Doesn't exist	Exists	Doesn't exist	Exists	Will do
Dashboard (My Comics)	Exists	Exists	N/A	Doesn't exist	Doesn't exist	Doesn't exist	Exists	Will do
Add new story	Exists	Exists	Manual	Manual	Exists	Exists	Doesn't exist	Will do
Edit story	Exists	Exists	Manual	Manual	Exists	Exists	Exists	Will do
Stories database / library	Exists	Exists	Chat history	Local files	Doesn't exist	Project files	Exists	Will do
Share to Facebook	Yes	Yes	Indirect	Manual	Manual	Manual	Exists	Will do (manual)
Share to Instagram	Export	Export	Indirect	Manual	Manual	Manual	Manual	Will do (manual)

3 Proposed solution

3.1 Solution Methodology

In line with best practice recommendations by related research, ComicMind will be designed to generate comics using a fixed 6-panel narrative structure to maintain coherence; implement fault-tolerant image generation so a single failure does not disrupt the workflow; allow retrying only the failed panels, improving control; use a minimal, clear interface to avoid cognitive overload; apply banned-word filtering to prevent low-quality or unsafe inputs; and provide a transparent, step-based flow (idea → story → images → preview → export).

Implementation Pipeline:

Feature: User Authentication

Frontend:

- Register and login forms
- Input validation
- “Remember Me” functionality using localStorage to store JWT tokens

Backend:

- Login and register routes
- Password hashing using bcrypt
- JWT token generation
- Middleware for protecting authenticated routes

Database:

- Users collection containing username, email, passwordHash, and createdAt

Feature: Comic Generation

Frontend:

- **Story input page where the user enters:**
 - Title
 - Story idea
 - Art style (e.g., anime, realistic, cartoon, noir)

- Generate button
- Loading indicator
- Comic preview page

Backend Pipeline:

1. Receive the title, idea, and selected style
2. Call the AI text model using a structured comic script prompt to generate a 6-panel story, including:
 - Panel descriptions
 - Dialogue
 - Narration
3. Construct image prompts using the panel descriptions and chosen art style
4. Generate images using a single AI image provider (OpenAI Images)
5. Return structured JSON containing all generated text and images
5. Return structured JSON containing all generated text and images

Database:

- Comics collection storing:
 - Title
 - Style
 - Panel descriptions
 - Dialogue and narration
 - Image URLs
 - Owner ID
 - createdAt

Feature: Comic Preview Page

Frontend:

- Display all 6 generated panels
- Overlay dialogue and narration boxes on top of each panel
- Two-page layout:

- Panels 1–3 on the left page
- Panels 4–6 on the right page
- Save Comic option
- Export options (PNG via html2canvas, PDF via jsPDF)

Backend:

- Endpoints for saving, loading, and deleting comics
- Ownership validation to ensure users only access their own comics

Feature: Gallery (My Comics):

Frontend:

- Display all comics created by the user
- Options to open, delete, or export each comic

Backend:

- Retrieve comics using page-by-page loading
- Validate that the requesting user owns the comic

Database:

- Stores essential comic data for retrieval and display

Feature: Content Safety:

Backend:

- Keyword filtering to prevent unsafe or inappropriate content before AI processing

Frontend:

- Display warnings when unsafe input is detected

General Implementation Steps (by Dependency Order)

- Set up user authentication: Build the login and registration pages, connect them to the Express backend, and create the users collection in MongoDB. This step ensures that all comics belong to registered users before any generation happens.
- Define the comic data model and build the main API routes: Create the MongoDB model to store the comic title, selected style, generated story text, the six panel images, and the user ID. Implement backend routes to create, save, fetch, and delete comics.

- Build the story-generation step using the OpenAI API: Implement a backend route that takes the user's title, style, and idea, sends them to the OpenAI text-generation endpoint, and returns a structured 6-panel story (panel descriptions, dialogue, and narration). This provides the full script before any images are generated.
- Add fault-tolerant image generation for all six panels: Integrate the OpenAI image generation API so that each of the six panel descriptions is processed independently. The system will attempt to generate all six images in a loop; if a single panel fails, it will not stop the entire process. Instead, failed panels are marked as "error," and only those panels can be retried after the initial generation.
- Create the comic preview layout: Build the two-page preview screen in React where all six panels appear with their generated text. Panels that successfully generated will show their images, while any failed panels will display a simple "retry" button.
- Build the "My Comics" gallery: Create the gallery page where logged-in users can view their saved comics. Each comic will have options to open, export, or delete it.
- Add export tools for PDF and PNG: Integrate html2canvas and jsPDF so the final comic layout can be exported as a single image or a multi-page PDF. Sharing to Instagram or Facebook will be handled manually by the user after export.
- Add a content-safety check using a banned-words file: Load a text file containing banned or unsafe words and check the user's input before sending anything to the OpenAI API. If the input contains restricted terms, the system displays a warning and blocks generation.

Implementation Technologies:

1. Frontend Technologies:

React:

- React is used to build the interface of the system. It makes it easy to create separate components for pages such as the story input screen, the comic preview, and the user gallery.
- React was chosen instead of Angular or Vue because it has a simpler setup and a larger community, which makes troubleshooting and learning easier.

TypeScript:

TypeScript adds type checking, which helps keep the code stable when handling structured data returned from the AI (panels, descriptions, dialogue, narration). It offers better reliability than plain JavaScript and reduces bugs when passing data between the frontend and backend.

Tailwind CSS:

Tailwind is used for styling because it allows fast layout development using utility classes. It is helpful for designing the comic pages, panel layouts, and speech bubble placements. It was chosen over frameworks like Bootstrap because it provides more flexibility and avoids predefined component limitations.

2. Backend Technologies:

Node.js:

Node.js runs the backend and is suitable for handling the AI requests and returning JSON responses. It works well with JavaScript/TypeScript, which keeps the project consistent across the frontend and backend. It was chosen instead of Python or PHP to avoid managing two different languages.

Express.js:

Express is used to create the backend routes for authentication, generating the story, generating images, and saving comics. It has a simple structure, which makes it easier to manage compared to larger frameworks like Nextjs.

JSON Web Tokens (JWT):

JWT handles user login and access control. It allows the system to secure routes without storing sessions on the server and It works well with modern frontend frameworks and was chosen for its simplicity.

3. Database:

MongoDB:

MongoDB is used to store comics, users, and panel data. Because MongoDB allows flexible JSON-like documents, it fits the structure of a comic (multiple panels, text, and images). A relational database like MySQL was not chosen because it requires fixed tables, which is harder to use for AI-generated content.

4. AI Integration

OpenAI API:

The OpenAI API is used to generate the story text and comic images. It was chosen because it produces structured story outputs and supports multiple art styles for the images. Other option like Stability ai was considered, but OpenAI provided more consistent results and easier integration with Node.js.

5. Export Tools

html2canvas and jsPDF

These tools allow the user to export the final comic as an image or a PDF directly from the browser.

They were chosen because they run entirely on the frontend and do not require any server-side processing.

ComicMind Resources:

Resources Created for the System:

1. Backend Components:

- User authentication (JWT login, registration, middleware)
- Story generation API route
- Image generation API route
- Comic saving and loading routes
- Database models for users and comics

2. Frontend Components:

- Story input page (title, idea, style)
- Comic preview page (panel layout, text overlays, export)
- My Comics gallery page
- Login and registration pages
- Export functions (PNG and PDF)

3. Data Structures and Logic:

- Structured 6-panel comic format

- Two-page comic layout
- Speech bubble and narration placement
- Page-by-page loading for gallery
- Simple content safety filter (keyword list)

4. UI and Styling:

- Panel layout grid
- Two-page comic view
- Bubble text components
- Tailwind CSS styling

Resources Borrowed (Integrated from External Sources):

1. AI Services

OpenAI API (text and image generation)

- Used to generate story panels, descriptions, dialogue, and narration
- Used to generate comic images in selected styles

2. Libraries and Tools

- React (frontend framework)
- TypeScript (type checking and structured development)
- Tailwind CSS (styling)
- Node.js + Express (backend routing)
- MongoDB (database for users and comics)
- html2canvas (comic export as PNG)
- jsPDF (comic export as PDF)

3.2 Functional/ Non-functional Requirements

Functional Requirements:

- User registration and login
- User dashboard to view saved comics
- Enter comic idea (title, style, description)
- Generate story phases (6-panel structure)
- Generate dialogue and narration
- AI image generation (single provider)
- Character-consistent image prompts
- Retry failed image panels
- SVG speech bubble placement
- Two-page comic preview
- Export comic to PNG
- Export comic to PDF
- Delete saved comics
- Content safety and banned-word filtering

Non-Functional Requirements:

- Fast performance for text and image generation
- Easy-to-use, simple, and intuitive user interface
- Secure password storage and validated user input
- System reliability with graceful error handling
- Basic readability (clear fonts, clean layout)
- Simple and clear step-by-step workflow

4 Conclusions and Changes

4.1 Summary

AI comic-generation tools still struggle to produce full, coherent comics because they focus mainly on generating single images rather than complete stories. Most existing platforms lack multi-panel continuity, consistent character appearances, and editable dialogue layers. Many tools also embed text directly into images or require users to rely on several paid services and manual editing, which slows down the process and makes it difficult for creators to build meaningful visual narratives. These limitations affect a wide range of users, especially non-artists, educators, and content creators who need an accessible way to turn ideas into structured and understandable comic stories.

ComicMind provides a solution by offering a unified system that generates a six-panel story, creates character-consistent images, and places editable speech bubbles without requiring additional tools. The platform transforms a simple user idea into organised story beats, generates clean images, and presents the final comic in a preview layout that can be saved or exported. Through its structured workflow, error handling, and user dashboard, ComicMind makes the entire process easier and more reliable. It allows users to create coherent, customisable comics quickly and without artistic skills, addressing the limitations found in existing AI comic-generation tools.

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